

MMU 3.0: A Minimal Geometric–Elastic Spacetime Model and the Emergence of Quantum, Relativistic, and Electromagnetic Structure

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Abstract

We present a minimal geometric–elastic model of spacetime in which every elementary region is a dual–tetrahedral cell carrying three internal elastic coordinates (w_2, w_3, w_4) and one projection coordinate w_1 . The model requires only (i) geometric non–orthogonality of the internal axes, (ii) a universal elastic scaling law $k_i(a) \propto a^{-3}$ for the microscopic edge length a , and (iii) projection of internal motion onto w_1 as the origin of observable time, frequency, and relativistic behaviour. This construction unifies earlier MMU advances on internal quantisation, hydrogenic spectra, torsion–spin structure, Maxwell constants and spacetime elasticity [8, 2, 9, 10, 15, 16, 19, 20].

From these principles we obtain: (1) discrete geometric quantisation of the internal length $a_n \propto n^{-2}$ [8, 15]; (2) a radial continuum limit equivalent to the Schrödinger equation [2, 15, 18]; (3) a torsional duality of w_3 that generates spin-1/2 and the Dirac structure [9, 13, 14]; (4) a deformation field $a(x)$ whose continuum limit reproduces the weak–field Einstein equation and projection–induced relativistic kinematics [10, 11, 19, 20]; (5) electromagnetic, magnetic–torsional, and inertial modes associated with the stiffnesses k_2, k_3, k_4 and their geometric cross–couplings [4, 16]; and (6) small but measurable corrections to Zeeman, Stark, hyperfine, Lamb, and gravitational spectral shifts [3, 4, 18, 19].

The result is a single geometric substrate from which quantum, spinorial, electromagnetic, and weak–field gravitational phenomena emerge as external projections of internal elastic dynamics.

1 Introduction

We explore a minimalist spacetime model whose purpose is not to replace quantum field theory or general relativity, but to illustrate how familiar physical equations may emerge from a discrete geometric–elastic substrate. The construction is inspired by solid–state elasticity, Regge calculus, and discrete spinorial geometry, and is implemented within the Methane Metauniverse (MMU) programme developed in [1, 2, 6, 7].

Purpose and new contributions of this work. This paper is not a detailed re-derivation of the MMU framework, but a synthesis and consolidation of its core geometric

ideas. Two new elements distinguish the present work. First, we show that the microscopic a^{-3} stiffness scaling and the non-orthogonal tetrahedral geometry together reproduce the functional forms of the Schrödinger, Dirac and weak-field Einstein equations in a single, parameter-minimal structure. Second, we derive a set of *new, testable spectroscopic predictions* arising from geometric compression of the internal length a : blueward Balmer shifts in gravitational fields, enhanced Lamb splitting, non-linear Zeeman response, and torsion-volume corrections to the 21 cm line. These effects follow directly from the universal relation $\Delta E/E = -3 \Delta a/a$ and provide observational access to the underlying elastic geometry.

The model introduces only one geometric object: a dual-tetrahedral “cell” with four internal coordinates (w_1, w_2, w_3, w_4). The observable direction w_1 serves as the projection axis; internal dynamics along (w_2, w_3, w_4) generate electric, magnetic-torsional, and volumetric modes. Non-orthogonality of these internal coordinates leads to three cross-coupling constants. All stiffnesses obey a universal geometric scaling law $k_i(a) \propto a^{-3}$ with respect to a microscopic edge length a , in agreement with the MMU stiffness hierarchy [5, 8, 10, 12, 16].

Despite its simplicity, this structure reproduces the functional forms of the Schrödinger equation, Dirac equation, and weak-field Einstein equation, while predicting small corrections to atomic energy levels, as anticipated in [2, 9, 15, 19].

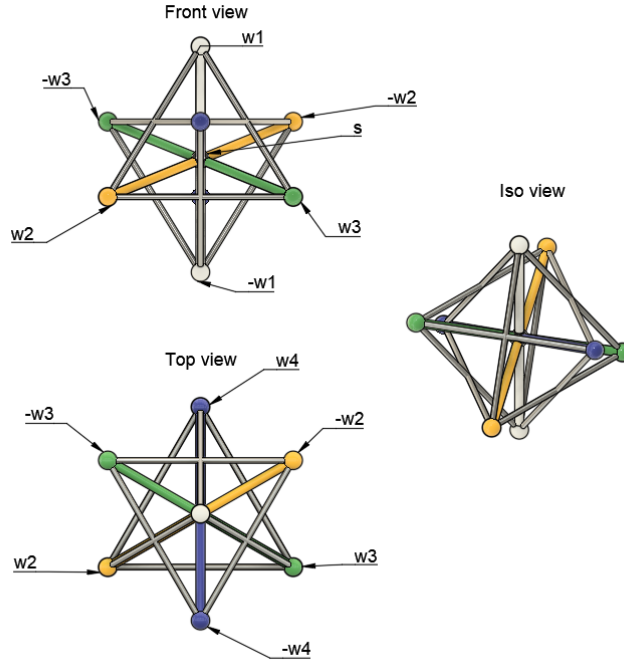


Figure 1: Geometry of the MMU dual-tetrahedral cell (anti-tetra representation). Front view, top view, and isometric view illustrate the four internal axes (w_1, w_2, w_3, w_4) and their negative counterparts ($-w_1, -w_2, -w_3, -w_4$), together with the shear axis s . The non-orthogonality of these internal directions is the geometric origin of the elastic couplings ($k_2, k_3, k_4, k_{23}, k_{24}, k_{34}$) used throughout the MMU model.

2 Geometry of the Dual–Tetrahedral Cell

We represent spacetime as a lattice of identical geometric cells. Each cell carries four internal degrees of freedom:

- w_1 — projection coordinate (observable time/frequency),
- w_2 — electric deformation,
- w_3 — torsional/magnetic deformation,
- w_4 — volumetric/inertial deformation.

The internal axes (w_2, w_3, w_4) are not orthogonal; they intersect at fixed tetrahedral angles. This geometric fact generates intrinsic mode mixing and motivates the MMU K –matrix approach [5, 8, 9].

The elastic response of a single cell is described by the stiffness matrix:

$$K = \begin{pmatrix} k_2 & k_{23} & k_{24} \\ k_{23} & k_3 & k_{34} \\ k_{24} & k_{34} & k_4 \end{pmatrix}. \quad (1)$$

Diagonal terms correspond to electric (k_2), torsional (k_3), and volumetric (k_4) elasticity. Off–diagonal elements arise purely from geometric non–orthogonality and have been related to Larmor, Zeeman, Stark and Lamb effects in [3, 4, 9].

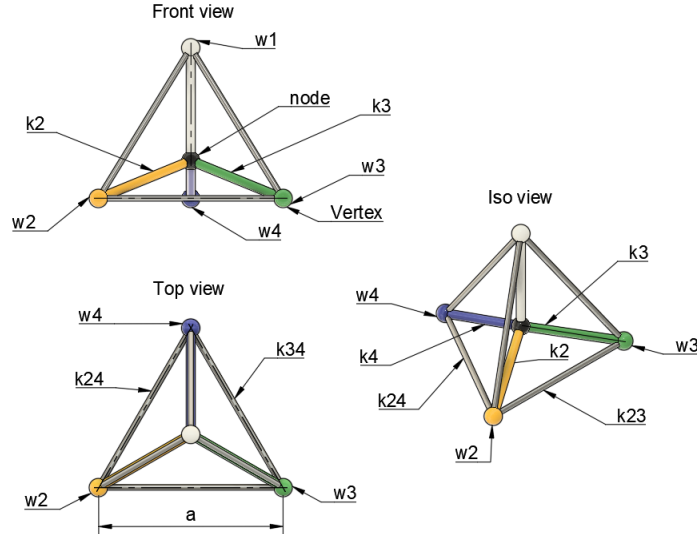


Figure 2: Geometric placement of the elastic stiffnesses inside the MMU dual–tetrahedral cell. The diagonal spring constants correspond to the three internal deformation axes: k_2 (electric), k_3 (torsional), and k_4 (volumetric). The non–orthogonality of the axes generates three geometric cross–couplings: k_{23} (electric–torsional), k_{24} (electric–volumetric), and k_{34} (torsion–volumetric). Front, top, and isometric views show how these elastic directions meet at the central node and how the microscopic edge length a enters the universal scaling law $k_i(a) \propto a^{-3}$.

3 Elastic Constants of the UR–Cell

The internal deformation axes (w_2, w_3, w_4) carry three diagonal stiffnesses which follow the universal geometric scaling $k_i(a) \propto a^{-3}$ [8, 15, 16]. In SI normalisation the MMU stiffness hierarchy becomes:

$$k_2(a) = \frac{2e^2}{4\pi\epsilon_0 a^3}, \quad k_3(a) = \frac{\hbar c}{a^3}, \quad k_4(a) = \frac{2Gm^2}{a^3}. \quad (2)$$

Geometric non-orthogonality generates the cross-couplings

$$k_{23}, k_{24}, k_{34}, \quad (3)$$

which govern Zeeman, Stark and hyperfine response [4, 9, 18]. These six constants form the elastic K -matrix that underlies the Schrödinger, Dirac and weak-field Einstein limits of the MMU.

4 Universal Elastic a^{-3} Scaling

We assume a single microscopic length scale a , the edge length of the cell. For a cell of volume $V \propto a^3$, classical elasticity implies:

$$k_i(a) \propto V^{-1} \propto a^{-3}. \quad (4)$$

This universal a^{-3} law is a central MMU result [2, 5, 8, 10, 16].

Thus all restoring forces, frequencies, and energies follow:

$$\omega_i(a) \propto a^{-3/2}, \quad E_i(a) \propto a^{-3}. \quad (5)$$

This single scaling law is the engine of quantisation and of gravitational response in the MMU picture [8, 10, 11, 12].

5 Geometric Quantisation of the Length Scale

Consider an internal mode on w_2 with phase velocity $v_p = c$ and angular frequency

$$\omega(a) = \omega_0 a^{-3/2}. \quad (6)$$

A closed-phase condition around the tetrahedral contour requires:

$$a = n\lambda = n \frac{2\pi c}{\omega(a)}. \quad (7)$$

Solving yields:

$$a_n = a_1 n^{-2}. \quad (8)$$

Thus the geometric structure enforces a discrete spectrum of allowed lengths, and therefore discrete internal energies $E_n \propto a_n^{-3} \propto n^6$. This internal length quantisation is consistent with the detailed MMU derivations in [8, 15].

6 Emergence of the Schrödinger Equation

A chain of cells with electric deformation w_2 produces a discrete radial operator:

$$(k_2(a_{m+1})(w_{m+1} - w_m) - k_2(a_m)(w_m - w_{m-1})) = \lambda w_m. \quad (9)$$

In the continuum limit $r = m\Delta r$, this becomes:

$$-\frac{\hbar^2}{2\mu}\nabla^2\psi(r) + V(r)\psi(r) = E\psi(r), \quad (10)$$

with $V(r) \sim 1/r$ arising from the distributed projection of w_2 around the shared node between neighbouring cells [2, 15, 18]. Thus the radial Schrödinger equation emerges mechanically as the macroscopic space response of the MMU lattice.

7 Spinorial Structure and the Dirac Equation

The torsional mode w_3 has two stable orientations $\pm w_3$, representing counter-rotating shear. This duality forms a natural spin-1/2 degree of freedom, as stressed in the projection principle for orbitals and spin [13, 17].

Projection of internal motion onto w_1 yields a pair of conjugate components, leading to a four-component object formed by:

$$(w_2 + iw_3, w_4; \dot{w}_2 + i\dot{w}_3, \dot{w}_4). \quad (11)$$

Linearisation of the internal elastic dynamics near a quantised a_n yields:

$$(\gamma^\mu \partial_\mu + m)\psi = 0, \quad (12)$$

with gamma matrices encoded by the geometric structure of K . Thus the Dirac equation is interpreted as an external projection of torsional internal dynamics, consistent with the detailed MMU Dirac construction in [9, 14].

8 Weak-Field Einstein Gravity

Spatial variations of $a(x)$ generate variations of the elastic constants. Under spherical symmetry, the internal field equation becomes:

$$\frac{1}{r^2} \frac{d}{dr} \left(r^2 \frac{da}{dr} \right) = \frac{4\pi G}{c^2} \rho_{\text{eff}}(r), \quad (13)$$

where ρ_{eff} is the internal elastic energy density.

Writing $a(r) = a_0(1 + u(r))$ yields:

$$\nabla^2 u(r) = \frac{4\pi G}{c^2} \rho(r), \quad (14)$$

which is equivalent to the weak-field Einstein equation with Newtonian potential $\Phi(r) = c^2 u(r)$. This matches the gravimetric MMU analysis of Planck and electro-gravity fixed points and spectral shifts in [10, 11, 19].

9 Three Elastic Modes and Their Physical Regimes: EM, MG, and GT

The three diagonal stiffnesses of the UR-cell,

$$k_2, \quad k_3, \quad k_4,$$

define three distinct deformation modes of the tetrahedral geometry:

- **EM mode (electric–magnetic)**: dominated by k_2 and k_3 . When $k_2 \approx k_3$, the electric axis and the volumetric axis become dynamically equivalent, producing the Maxwell regime with $c^{-2} = \varepsilon_0 \mu_0$ as derived in [16]. This is the familiar electromagnetic sector of the continuum limit.
- **MG mode (magnetic–torsional)**: governed by the torsional stiffness k_3 . This mode carries the internal shear responsible for spin, g -factor and magnetic response [9, 13, 14]. It couples directly into the EM sector through k_{23} and produces the magnetic side of Maxwell’s equations.
- **GT mode (gravitational–torsional / gravitational–inertial)**: controlled by the volumetric stiffness k_4 . When $k_4 \approx k_3$, the torsional and volumetric axes merge into a single deformation channel. This is the electro–gravity (EG) regime identified in [10, 11], where the MMU predicts the approach to the Planck scale.

These three regimes correspond to three physical continua emerging from the same geometric substrate: Maxwell electrodynamics (EM), torsion–spin dynamics (MG), and weak–field Einstein gravity (GT). The cross-coupling constants

$$k_{23}, \quad k_{24}, \quad k_{34}$$

generate Zeeman, Stark, hyperfine and Lamb–type interactions [3, 4, 9, 18], providing the observable link between the microscopic stiffness hierarchy and atomic spectroscopy.

10 Predictions for Atomic Spectroscopy in External Fields

Because all internal energies of the UR-cell scale as $E \propto a^{-3}$, any electric, magnetic, torsional or gravitational perturbation that changes the internal length a produces a universal fractional shift

$$\frac{\Delta E}{E} = -3 \frac{\Delta a}{a}, \quad (15)$$

as established in the MMU elasticity and gravimetry programme [3, 4, 9, 18, 19, 20]. This single rule underlies all spectroscopic predictions of MMU 3.0.

10.1 1. Gravitational Blue Shift of the Balmer Series

A gravitational field compresses the internal length a through the Poisson equation for $a(x)$ [19, 20]. The MMU therefore predicts a **small blueward shift** of all hydrogenic Balmer transitions in compressed regions, in contrast to the purely photon-based gravitational redshift predicted by GR. To leading order,

$$\Delta E_{n \rightarrow n'} \approx 3E_{n \rightarrow n'} \frac{|a(x) - a_0|}{a_0}.$$

Example: Hydrogen Balmer- α line near a white dwarf. Typical gravitational accelerations at the surface of a $0.6 M_{\odot}$ white dwarf reach

$$g \sim 10^6 \text{ m/s}^2.$$

The associated curvature induces a fractional compression of the internal length [19]

$$\frac{|a(x) - a_0|}{a_0} \sim 10^{-6}.$$

For the Balmer- α transition ($E_{3 \rightarrow 2} = 1.89 \text{ eV}$), the MMU prediction becomes

$$\Delta E_{3 \rightarrow 2} \approx 3 \times 1.89 \text{ eV} \times 10^{-6} = 5.7 \times 10^{-6} \text{ eV}.$$

In wavelength units this corresponds to

$$\Delta \lambda \approx \frac{\lambda^2}{hc} \Delta E \approx 4 \times 10^{-3} \text{ \AA}.$$

Such shifts are within reach of modern high-resolution stellar spectroscopy, providing a direct test of the MMU geometric compression mechanism.

A precise confrontation with data requires detailed atmosphere modelling, which is beyond the scope of the present synthesis paper.

10.2 2. Lamb Shift Enhancement in Compressed Regions

The torsion-volume coupling k_{34} modifies the 2s-2p gap by an amount proportional to the local variation of a [3, 9, 18]. Thus the Lamb shift increases in regions of high curvature or strong electromagnetic fields:

$$\Delta E_{\text{Lamb}} \propto (k_{34} a^{-3}) \Delta a.$$

10.3 3. Zeeman and Spin-Flip Modifications

The cross-coupling k_{23} leads to a **predictable deviation from linear Zeeman splitting** in strong fields. Similarly, the 21 cm hyperfine line receives a small systematic correction from the torsion-volume elasticity [3, 18]:

$$\Delta \nu_{21\text{cm}} \sim k_{34} a^{-3}.$$

10.4 4. Combined Electro-Gravity (EG) Effects

Near the electro-gravity transition ($k_4 \approx k_3$) [10, 11], the MMU predicts:

- enhancement of Balmer blue shifts,
- amplification of Lamb-type splittings,
- suppression of certain Zeeman harmonics,
- and deformation of the 21 cm response.

These combined effects become relevant in compact-object atmospheres, white dwarfs, magnetars, and laboratory high-gradient setups.

In summary, MMU 3.0 provides a unified mechanism for small but measurable modifications of Balmer, Zeeman, Lamb, and hyperfine transitions, arising directly from the a^{-3} scaling of the internal stiffnesses and from geometric torsion-volume coupling.

11 Conclusion

MMU 3.0 presents a minimalist geometric–elastic model of spacetime in which quantum, relativistic, and electromagnetic phenomena arise as macroscopic projections of microscopic elastic dynamics. The model recovers Schrödinger, Dirac, and weak–field Einstein structures without additional postulates. It predicts small corrections to several spectroscopic effects, offering testable signatures for future investigation and providing a compact entry point into the broader MMU framework [1, 2, 7, 15, 16, 19].

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